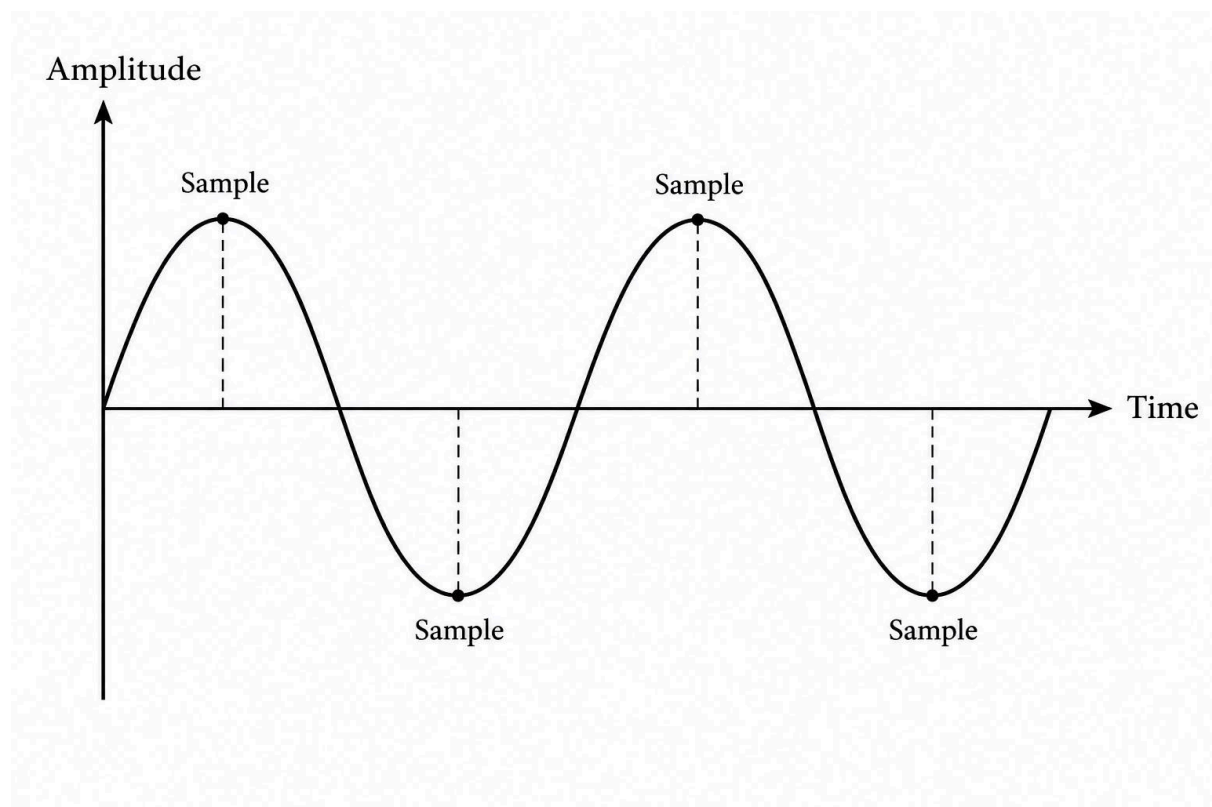


PCM (Pulse Code Modulation)

PCM is a method used to convert an analog signal into a digital format in three sequential steps: Sampling, Quantizing and Encoding

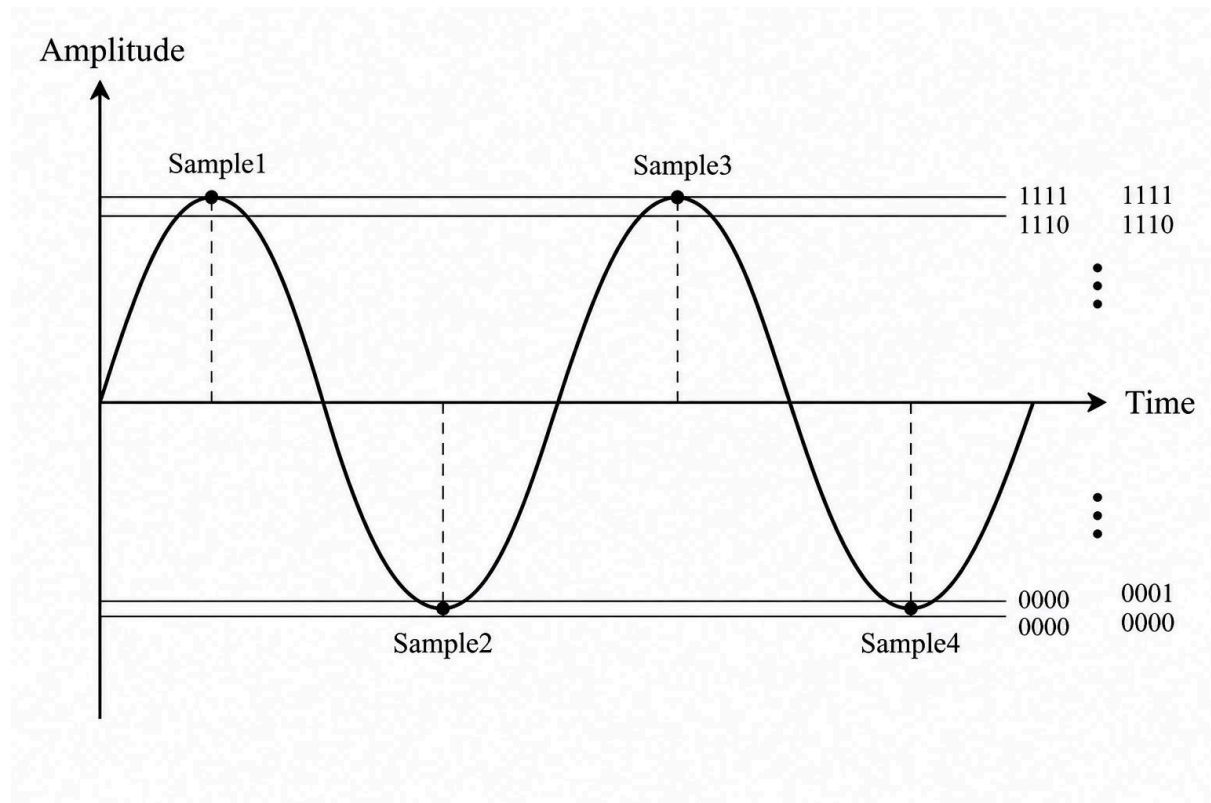
The examples below use only two sine waves for simplicity. In voice communication, speech signals typically contain frequencies up to approximately 4,000 Hz.

1) Sampling is the process of measuring the amplitude of a continuous analog audio wave at regular time intervals, which can be visualized as drawing vertical lines from the time axis to touch the continuous signal curve.



According to the **Nyquist-Shannon sampling theorem**, the sampling rate (f_s) must be at least twice the maximum frequency (f_{max}) of the analog signal ($f_s \geq 2 \cdot f_{max}$); otherwise, **aliasing** occurs, causing high-frequency components to overlap as low-frequency artifacts and permanently distorting the reconstructed signal.

2)Quantizing is the process of mapping continuous sampled amplitudes to the nearest fixed discrete levels, visualized as horizontal grid lines along the amplitude axis. The resolution of these grid lines is defined by the **bit depth**, which determines how many discrete levels are available (such as 8-bit with 256 levels, 16-bit with 65,536 levels, or 24-bit with over 16.7 million levels).



3)Encoding Encoding is the final stage of PCM, in which each quantized amplitude value is represented by a corresponding binary code consisting of 0s and 1s. These binary codes are then arranged sequentially according to the order of the samples, from the first time sample to the last, forming a continuous digital bitstream.

Encoded output for the example above: 1111 1111 0000 0000 1111 1111 0000 0000

In voice communication, a data rate of approximately 64,000 bit per second is commonly used for uncompressed PCM audio. $f_s \cdot \text{bit depth} \cdot \text{channel count}$)

Key Concepts in Encoding

Binary Mapping: Each horizontal quantization level assigned to a sample is expressed as an N-bit binary number determined by the bit depth.

Sequential Stream: The converted binary numbers are arranged back-to-back in chronological order (left-to-right along the timeline: Sample 1, Sample 2, Sample 3, ...) resulting in a unified line of raw audio data.